

SSH Tunneling and Network Traffic Validation

Professional Case Report 15

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Case ID	CSC-2026-15
Discipline	Network security
Overall risk	High
Evidence status	Laboratory methodology documented

Purpose

Explain how tunneling changes traffic flow and show how defenders can validate the actual connection path.

Scope note

This report documents work performed in an authorized training environment. Confirmed findings are separated from methodology and expected laboratory behavior.

Executive Summary

A Windows-to-Linux SSH connection and alternate-port forwarding scenario was observed with packet capture to distinguish normal and tunneled traffic. The exercise showed that port numbers do not establish protocol identity. Interface-aware packet capture is necessary to understand forwarding and tunnel behavior.

Case Profile

Scenario	A Windows-to-Linux SSH connection and alternate-port forwarding scenario was observed with packet capture to distinguish normal and tunneled traffic.
Environment	Windows client, Kali Linux host and isolated lab network with TCP 22 and 4444 observations.
Tools	OpenSSH, Netcat, tcpdump, Windows command line

Methodology

I used the following sequence to keep the analysis repeatable and to separate observation from interpretation.

1. Established a normal SSH session for a baseline.
2. Tested alternate-port connectivity and identified forwarding dependencies.
3. Captured traffic on the external interface and loopback interface.
4. Compared source, destination and port observations.
5. Closed listeners and sessions after validation.

Technical Findings

ID	Finding	Risk	Evidence and analysis
F01	Alternate-port tunnel	High	Traffic on port 4444 can carry another protocol and should not be trusted based only on port number.
F02	Interface-specific evidence	Medium	External and loopback captures reveal different stages of forwarded traffic.
F03	Encrypted content	Medium	Network monitoring may identify endpoints and timing even when payload content is protected.

Representative Commands and Actions

The following commands or analysis actions represent the core of the practical workflow.

```
ssh msfadmin@<linux_ip>
ssh msfadmin@<linux_ip> -p 4444
tcpdump -nn -i eth0 port 4444
tcpdump -nn -i lo port 22
```

Risk Assessment

Risk	Level	Operational impact
Covert channel	High	Tunneling can bypass simplistic port-based controls
Unauthorized remote access	High	Forwarded services may expose management interfaces

Recommendations

1. Restrict SSH access to managed sources.
2. Monitor unusual destination ports carrying SSH-like traffic.
3. Use application-aware controls instead of port-only policies.
4. Review authorized keys and forwarding settings regularly.

Knowledge Demonstrated

- SSH tunneling
- Packet capture
- Traffic-flow analysis
- Control validation

Conclusion

The exercise showed that port numbers do not establish protocol identity. Interface-aware packet capture is necessary to understand forwarding and tunnel behavior.

Hands On Lab Practical 15

Completed as part of the authorized cyber security course practical work.